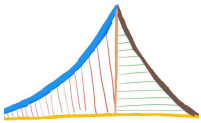
Math for EE
GATE CS1 2025



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE CS1 2025

A. Formulae

1.1.1 Probability Mass Function (PMF) and Z-Transform of Discrete Random Variables:

$$M_Y(Z) = E[Z^{-Y}] = \sum_k P_Y(k) Z^{-k} \tag{1.1.1.1}$$

1.1.2 Z-Transform of Sum of Independent Random Variables $Y = \sum_{i=1}^{n} Y_i$:

$$M_Y(Z) = \prod_{i=1}^{n} M_{Y_i}(Z) \tag{1.1.2.1}$$

1.1.3 Combinations with Repetition (Stars and Bars Principle):

$$N = \binom{n+r-1}{r} \tag{1.1.3.1}$$

1.1.4 Composition of Functions and Surjectivity (Onto Property):

$$(f \circ g)(x) = f(g(x)), \quad \forall x \in A \tag{1.1.4.1}$$

1.1.5 Shortest Path Inequality in Subgraphs and Spanning Trees: If T is a minimum spanning tree of graph G, then for any vertices $u, v \in V(G)$:

$$d_1(u, v) \leq d_2(u, v) \tag{1.1.5.1}$$

where $d_1(u, v)$ is the shortest path in G and $d_2(u, v)$ is the shortest path in T.

1.1.6 Recurrence Relation Reduction using Exponential Division:

$$T(n) = 2T(n-1) + n2^n \implies \frac{T(n)}{2^n} = \frac{T(n-1)}{2^{n-1}} + n \implies T(n) = \Theta(n^2 2^n) \tag{1.1.6.1}$$

1.1.7 Determinant Conditions for Linear System Consistency $\mathbf{Ax} = \mathbf{b}$:

$$\det(\mathbf{A}) \neq 0 \iff \text{Unique Solution} \tag{1.1.7.1}$$

1.1.8 Majority Boolean Function Representation:

$$X(a, b, c) = ab + bc + ca \tag{1.1.8.1}$$

1.1.9 Sign Extension Rule in 2's Complement Numbers: To extend an n-bit signed integer to an m-bit integer ($m > n$), replicate the Most Significant Bit (MSB / sign bit) across all added higher-order bit positions.

1.1.10 Number of Strings over Alphabet Size k without Adjacent Duplicate Symbols:

$$N_{\text{no_consec}} = k(k-1)^{n-1} \tag{1.1.10.1}$$

1.1.11 Boundary Differentiability Condition for Piecewise Functions:

$$\lim_{x \to x_0^-} f'(x) = \lim_{x \to x_0^+} f'(x) \tag{1.1.11.1}$$

1.1.12 Bayes' Theorem for Conditional Probability:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{1.1.12.1}$$

1.1.13 Total Inversions (Swaps) in a Reverse-Sorted Array of Size n:

$$\text{Swaps} = \frac{n(n-1)}{2} \tag{1.1.13.1}$$

1.1.14 Eigenvalues of Matrix Powers: If λ is an eigenvalue of matrix $\mathbf{A}$, then λ^k is an eigenvalue of $\mathbf{A}^k$:

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \implies \mathbf{A}^k\mathbf{x} = \lambda^k\mathbf{x} \tag{1.1.14.1}$$

1.1.15 Minimum Height of Breadth-First-Search (BFS) Tree given Graph Diameter D:

$$h(T) \geq \left\lceil \frac{D}{2} \right\rceil \tag{1.1.15.1}$$

1.1.16 Algebraic Monoid Structure under Function Addition: A set of non-negative integer functions under pointwise addition $(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$ forms an Abelian monoid with identity element $e(n) = 0$.

1.1.17 Basic Block Leader Criteria in Compiler Intermediate Code:

$$\text{Leaders} = \{\text{First Instruction}, \text{Targets of Jumps}, \text{Instructions Immediately Following Jumps}\} \tag{1.1.17.1}$$

1.1.18 Independent Bernoulli Trials for n-bit Data Transmission:

$$P(\text{Error-Free}) = (1-p)^n \tag{1.1.18.1}$$

where p is the bit-flip probability per bit.

1.1.19 Continuous Probability Density Function (PDF) Normalization:

$$\int_{-\infty}^{\infty} P(x)\,dx = 1, \quad P(a \leq x \leq b) = \int_a^b P(x)\,dx \tag{1.1.19.1}$$

1.1.20 State Equivalence Reduction in Finite State Machines (FSM): Two states S_i and S_j are equivalent if for all input symbols x:

$$\text{Out}(S_i, x) = \text{Out}(S_j, x) \quad \text{and} \quad \text{Next}(S_i, x) \equiv \text{Next}(S_j, x) \tag{1.1.20.1}$$

1.1.21 Period of an n-bit Johnson / Twisted-Ring Counter: An n-bit Johnson counter constructed with inverted feedback ($D_0 = \overline{Q_{n-1}}$) cycles through $2n$ distinct state transitions:

$$N_{\text{states}} = 2n \tag{1.1.21.1}$$

1.1.22 Cycle Property of Minimum Spanning Trees (MST): For any simple cycle C in graph G, the strictly maximum-weight edge in C cannot belong to any MST of G.

1.1.23 Open Addressing Double Hashing Probe Sequence:

$$h(k, i) = [h_1(k) + i \cdot h_2(k)] \bmod m \tag{1.1.23.1}$$

where $i \in \{0, 1, \ldots, m-1\}$ represents the probe iteration count and m is the hash table capacity.

B. Problems

1.2.1 A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once? (CS1 2025)

a) $\frac{75}{216}$ b) $\frac{1}{6}$ c) $\frac{1}{18}$ d) $\frac{25}{216}$

1.2.2 A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase? (CS1 2025)

a) 4 b) 20 c) 24 d) 48

1.2.3 $g(\cdot)$ is a function from A to B, $f(\cdot)$ is a function from B to C, and their composition defined as $f(g(\cdot))$ is a mapping from A to C. If $f(\cdot)$ and $f(g(\cdot))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(\cdot)$? (CS1 2025)

a) $g(\cdot)$ must be an onto (surjective) function.
b) $g(\cdot)$ must be a one-to-one (injective) function.
c) $g(\cdot)$ must be a bijective function, that is, both one-to-one and onto.
d) $g(\cdot)$ is not required to be a one-to-one or onto function.

1.2.4 Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let $d_1(u, v)$ and $d_2(u, v)$ be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v? (CS1 2025)

a) $d_1(u, v) = d_2(u, v)$
b) $d_1(u, v) \leq d_2(u, v)$
c) $d_1(u, v) \geq d_2(u, v)$
d) $d_1(u, v) \neq d_2(u, v)$

1.2.5 Consider the following recurrence relation: $T(n) = 2T(n-1) + n2^n$ for $n > 0$, $T(0) = 1$. Which ONE of the following options is CORRECT? (CS1 2025)

a) $T(n) = \Theta(n^2 2^n)$
b) $T(n) = \Theta(n2^n)$
c) $T(n) = \Theta((\log n)^2 2^n)$
d) $T(n) = \Theta(4^n)$

1.2.6 Consider the given system of linear equations for variables x and y, where k is a real-valued constant:

$$x + ky = 1$$
$$kx + y = -1$$

Which of the following option(s) is/are CORRECT? (CS1 2025)

a) There is exactly one value of k for which the above system of equations has no solution.
b) There exist an infinite number of values of k for which the system of equations has no solution.
c) There exists exactly one value of k for which the system of equations has exactly one solution.
d) There exists exactly one value of k for which the system of equations has an infinite number of solutions.

1.2.7 Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables? (CS1 2025)

a) $X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$
b) $X(a, b, X(a, b, c)) = X(a, b, c)$
c) $X(a, b, X(a, c, d)) = (X(a, b, a)$ AND $X(c, d, c))$
d) $X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

1.2.8 The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of -6? (CS1 2025)

a) 1000 1010 in 8-bits
b) 1111 1010 in 8-bits
c) 1000 0000 0000 1010 in 16-bits
d) 1111 1111 1111 1010 in 16-bits

1.2.9 Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, "aa", "bb" or "cc". The number of such strings of length 5 that are possible is _____. (CS1 2025)

1.2.10 Consider the given function $f(x)$:

$$f(x) = \begin{cases} ax + b, & \text{for } x < 1 \\ x^3 + x^2 + 1, & \text{for } x \geq 1 \end{cases}$$

If the function is differentiable everywhere, the value of b must be _____. (CS1 2025)

1.2.11 A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(\text{head}) = 0.5$ and for a fake coin, $P(\text{head}) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (CS1 2025)

1.2.12 The pseudocode of a function `fun()` is given below:

```
fun (int A[0,...,n−1]) {
    for i = 0 to n−2
        for j = 0 to n−i−2
            if (A[j] > A[j+1])
                then swap A[j] and A[j+1]
}
```

Let $A[0, \ldots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function `fun()` is called with $A[0, \ldots, 29]$ as argument, is _____. (CS1 2025)

1.2.13 Consider the following C program:

```
#include <stdio.h>

void foo (int *p, int x) {
    *p = x;
}

int main() {
    int *z;
    int a = 20, b = 25;
    z = &a;
    foo (z, b);
    printf ("%d", a);
    return 0;
}
```

The output of the given C program is _____. (CS1 2025)

1.2.14 Let A be a 2×2 matrix as given:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13}? (CS1 2025)

a) $1, -1$
b) $2\sqrt{2}, -2\sqrt{2}$
c) $4\sqrt{2}, -4\sqrt{2}$
d) $64\sqrt{2}, -64\sqrt{2}$

1.2.15 Consider the following four variable Boolean function in sum-of-product form:

$$F(b_3, b_2, b_1, b_0) = \Sigma(0, 2, 4, 8, 10, 11, 12)$$

where the value of the function is computed by considering $b_3 b_2 b_1 b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F? (CS1 2025)

a) $\overline{b}_1\overline{b}_0 + \overline{b}_2\overline{b}_0 + b_1\overline{b}_2 b_3$
b) $\overline{b}_1\overline{b}_0 + \overline{b}_2\overline{b}_0$
c) $\overline{b}_2\overline{b}_0 + b_1 b_2 b_3$
d) $\overline{b}_0\overline{b}_2 + \overline{b}_3$

1.2.16 Let $G(V, E)$ be an undirected and unweighted graph with 100 vertices. Let $d(u, v)$ denote the number of edges in a shortest path between vertices u and v in V. Let the maximum value of $d(u, v)$, $u, v \in V$ such that $u \neq v$ be 30. Let T be any breadth-first-search tree of G. Which ONE of the given options is CORRECT for every such graph G? (CS1 2025)

a) The height of T is exactly 15.
b) The height of T is exactly 30.
c) The height of T is at least 15.
d) The height of T is at least 30.

1.2.17 $A = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in F$ we define

$$(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$$

for every number n in A. Which of the following is/are CORRECT about the mathematical structure $(F, \odot)$? (CS1 2025)

a) $(F, \odot)$ is an Abelian group.
b) $(F, \odot)$ is an Abelian monoid.
c) $(F, \odot)$ is a non-Abelian group.
d) $(F, \odot)$ is a non-Abelian monoid.

1.2.18 Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is _____. (CS1 2025)

```
1001: i = 1
1002: j = 1
1003: t1 = 10 * i
1004: t2 = t1 + j
1005: t3 = 8 * t2
1006: t4 = t3 - 88
1007: a[t4] = 0.0
1008: j = j + 1
1009: if j <= 10 goto 1003
1010: i = i + 1
1011: if i <= 10 goto 1002
1012: i = 1
1013: t5 = i - 1
1014: t6 = 88 * t5
1015: a[t6] = 1.0
1016: i = i + 1
1017: if i <= 10 goto 1013
```

1.2.19 Consider the following database tables of a sports league:

- `player (pid, pname, age)`
- `coach (cid, cname)`
- `team (tid, tname, city, cid)`
- `members (pid, tid)`

An instance of the table and an SQL query are given below.

player

pid	pname	age
1	Jasprit	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
103	Trevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

members

pid	tid
1	10
2	30
3	10
4	20

```
SELECT MIN (P.age)
FROM player P
WHERE P.pid IN (
    SELECT M.pid
    FROM team T, coach C, members M
    WHERE C.cname = 'Mark'
      AND T.cid = C.cid
      AND M.tid = T.tid
)
```

The value returned by the given SQL query is _____. (CS1 2025)

1.2.20 Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____. (CS1 2025)

1.2.21 Consider a probability distribution given by the density function $P(x)$:

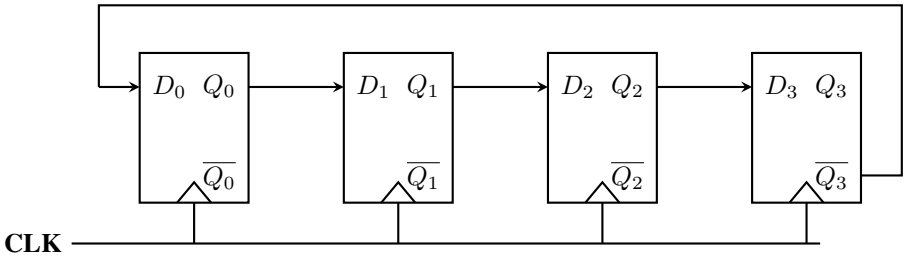
$$P(x) = \begin{cases} Cx^2, & \text{for } 1 \leq x \leq 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$

The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____. (CS1 2025)

1.2.22 Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (CS1 2025)

Present state	Next state		Output f	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

1.2.23 Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____. (CS1 2025)



1.2.24 Consider the following C program:

```
#include <stdio.h>

int foo (int S[], int size) {
    if (size == 0) return 0;
    if (size == 1) return 1;
    if (S[0] != S[1]) return 1 + foo(S + 1, size − 1);
    return foo(S + 1, size − 1);
}

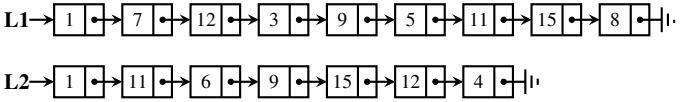
int main() {
    int A[] = {0, 1, 2, 2, 2, 0, 0, 1, 1};
    printf ("%d", foo(A, 9));
    return 0;
}
```

The value printed by the given C program is _____. (CS1 2025)

1.2.25 Let `LIST` be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
    int data;
    struct list *next;
} LIST;
```

Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes.



Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____. (CS1 2025)

```
int find (int query, LIST *list) {
    while (list != NULL) {
        if (list->data == query) return 1;
        list = list->next;
    }
    return 0;
}

int main () {
    ptr1 = L1; ptr2 = L2;
    while (ptr1->next != NULL) {
        query = ptr1->next->data;
        if (find (query, L2))
            ptr1->next = ptr1->next->next;
        else
            ptr1 = ptr1->next;
    }
    return 0;
}
```

1.2.26 Consider the following C program:

```
#include <stdio.h>

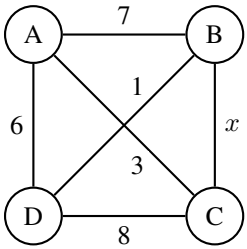
int gate (int n) {
    int d, t, newnum, turn;
    newnum = turn = 0; t = 1;
    while (n >= t) t *= 10;
    t /= 10;
    while (t > 0) {
        d = n / t;
        n = n % t;
        t /= 10;
```

```
        if (turn) newnum = 10 * newnum + d;
        turn = (turn + 1) % 2;
    }
    return newnum;
}

int main () {
    printf ("%d", gate (14362));
    return 0;
}
```

The value printed by the given C program is _____. (CS1 2025)

1.2.27 The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____. (CS1 2025)



1.2.28 In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i-th probe in the open address table is $[h_1(k) + i \cdot h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____. (CS1 2025)